Asset Management Challenge 2026

First round: the Strategic Asset Allocation

Financial Modeling, USI Lugano. These guidelines complete the general guidelines of the challenge.

1. The challenge in three rounds

- **Round 1.** Each team chooses its own **Strategic Asset Allocation (SAA)**. There is no common benchmark.
- **Rounds 2 and 3.** The teams deviate tactically from their SAA, with the strategies of the second part of the course, within a tracking-error budget. The SAA itself does not change: it is the reference of the team until the end of the challenge.
- In all rounds, performance is measured by the **Sharpe Ratio** of the portfolio.

2. Key dates

Thursday 15 October	Send the link to the website of your fund and the logo by e-mail to the teaching assistant
Saturday 17 and Sunday 18 October	Insert the positions in the Bloomberg portfolio. Bloomberg uses the closing prices of Friday 16 October
Sunday 18 October, 23:59	Deadline for the report and the portfolio description (all groups). By the same time, the Bloomberg portfolio is shared with the teaching assistant
Tuesday 20 October, 11:00	Presentations of groups 1, 2, 3, 4 and 5 (room SI-006)
Thursday 22 October, 13:30	Presentations of the other groups (Bloomberg Lab, SI-008)
Until the second round	The portfolio remains unchanged until the second round (7 and 8 November)

3. The task

Each team allocates an initial capital of **USD 100 million** to its Strategic Asset Allocation: the long-run allocation across asset classes that the team considers most appropriate, given what we discussed in Class 3 (Parts I and II).

- The portfolio can contain only **long positions in ETFs** that track broad asset classes, such as equities, fixed income, and commodities. Single stocks, short positions, leverage, and inverse or leveraged ETFs are not admitted in this round.
- Within this universe, you are free to choose the asset classes, the sectors, and the geographical areas.
- Keep the number of positions low.
- You can keep part of the portfolio in cash (USD). Cash does not earn interest.
- You choose the method that determines the weights: constrained mean-variance optimization, minimum variance, equal weight, risk parity, or another rule. The report must justify the choice.

4. The risk budget

The **target volatility of the SAA is 8% per year**. The ex ante volatility of the portfolio must lie **between 7% and 9%**.

- **Ex ante volatility.** Compute it from the covariance matrix of the monthly returns of the ETFs in USD, annualized (multiply the monthly volatility by $\sqrt{12}$). Use the last 60 months where they are available. If an ETF has a shorter history, use the index that it tracks, or a comparable ETF, for the missing months, and state it in the report.
- **Value at Risk.** Translate the risk of the portfolio into a Value at Risk: report the one-month VaR at the 5% level, as a percentage and in USD. As a reference, under the normal distribution a volatility of 8% per year corresponds to a one-month 5% VaR of about 3.8% of the portfolio, that is, USD 3.8 million.

The risk budget applies to the SAA. The tactical deviations of rounds 2 and 3 have their own budget, stated in the guidelines of those rounds.

5. Evaluation

The performance of the portfolio is measured by the **Sharpe Ratio**:

$$SR = \frac{\bar{R}_P - \bar{R}_f}{\sigma(R_P)}$$

where $\bar{R}_P$ is the average return of the portfolio, $\bar{R}_f$ is the average risk-free rate (the 1-month US Treasury bill rate), and $\sigma(R_P)$ is the standard deviation of the return of the portfolio. The Sharpe Ratio is computed on the returns realized during the challenge. We also compare the realized volatility of the portfolio with the target.

6. The backtest

- Backtest the allocation with **monthly rebalancing** to the weights of the SAA.
- You can download adjusted prices from Bloomberg (with the Excel add-in on the terminals of the Bloomberg Lab) or from WRDS (write to katia.mue@usi.ch to activate an account).
- You are free to use Python, Excel, R, or another language.

7. The report

The report contains two parts: the analysis of the allocation, and the checkpoint assignments given in class.

Analysis of the allocation

1. The universe of assets and the reasons for its choice.
2. The inputs: the volatilities and the correlations, the expected returns if the method uses them, and the source of each.
3. The method that determines the weights, and the reasons for its choice.
4. The allocation: for each position, the capital weight and the risk contribution. For the portfolio, the ex ante volatility and the VaR.
5. The backtest: how it was run, and the historical performance and risk of the allocation, with the measures of Class 2 (among them the Sharpe Ratio and the maximum drawdown). Include a plot of the cumulative returns.
6. The code, attached to the report, with a short description of what each file does.

7. The portfolio description: for each position, the name of the security, the identifier (Bloomberg ticker or ISIN), the currency, the number of shares, the weight, and the date of the prices used.

Checkpoint assignments

The report must include the three checkpoint assignments given in class. Each is graded as complete or incomplete, together with the report. The full text of each assignment is on the last slides of the corresponding in-class deck on iCorsi.

1. **The investment policy (Class 3, Part I).** One page, placed as the first page of the report: the universe; the objective and the risk budget (the volatility target, the VaR, and the maximum acceptable drawdown); the constraints that the team imposes on itself, each with its reason; the capital weight and the risk contribution of each position.
2. **The comparison with the robust approaches (Class 3, Part II).** Half a page: the weights of your SAA next to the weights under equal weight and under risk parity, each with its risk contributions and its volatility, and one paragraph on where the rules disagree with your weights and what justifies your choice.
3. **The backtest protocol (Class 4).** One section: the design of the backtest; which part of the data was used to choose the allocation and which was held out, and how many variations the team tried; the volatility, the maximum drawdown, and the 5% VaR of the backtested allocation; the backtested Sharpe Ratio, the discounted value that the team considers realistic, and the reason for the discount.

Delivery

Send the report, the code, and the portfolio description by e-mail to the teaching assistant (riccardo.croci@usi.ch) by **Sunday 18 October, 23:59**. The deadline is the same for all groups, whatever the day of the presentation. Upload the same material to the website of your fund.

8. The portfolio in Bloomberg

The performance is tracked with a Bloomberg portfolio, which each team creates.

1. Create the portfolio with the function `PRTU`. When asked for the type of portfolio, choose *equity* and *shares/par amount*.
2. Enter the positions as numbers of shares. Convert the weights into numbers of shares with the closing prices of Friday 16 October and the capital of USD 100 million.
3. After creating the portfolio, select *Action* in the red toolbar, then *Update Corporate Actions*, and tick both "adjust cash for coupon payments and dividends" and "reinvest cash dividends". With these options, the number of shares is fixed at inception, the value of the portfolio follows the prices every day, and cash dividends are reinvested in the security that pays them.
4. Insert the positions on **Saturday 17 or Sunday 18 October**, so that Bloomberg uses the closing prices of Friday 16 October.
5. Share the portfolio with the Bloomberg account of the teaching assistant (riccardo.croci@usi.ch). The option is in the red toolbar of the *Portfolio Administration* screen. We use it to monitor the performance and to verify that the portfolio corresponds to the one in the report.

The portfolio in Bloomberg must be identical to the portfolio in the report, and it cannot be changed before the second round.

9. The presentation

- Each group has about **15 minutes**. The list of the groups is on iCorsi.
- All the members of the group must be present and take part in the presentation.
- Present yourselves as a professional team, with a name for your fund.
- Start with a market outlook that is consistent with the asset classes and the geographical areas of the portfolio. Then describe the allocation, the reasons for it, and the results of the backtest.
- The presentation is also a pitch: the other groups are your potential investors.
- During the presentations of the other groups, you are expected to perform due diligence: ask questions and challenge their choices.